An Explainable Action-Token Generation Framework Based on Bladder Ultrasound Segmentation and Image Quality Assessment

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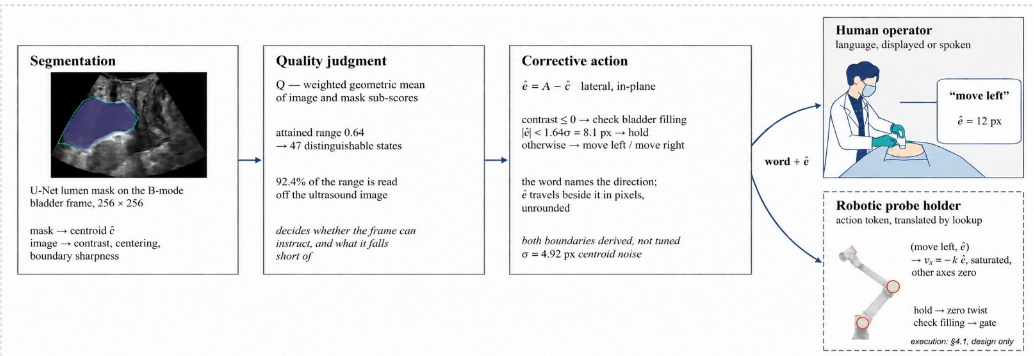
Abstract: Automated ultrasound-image analysis requires an explicit, auditable representation mapping image-derived states to actionable structured outputs. We convert bladder ultrasound segmentation and image-quality assessment into a safety-constrained, machine-readable action token $\tau = (a, \hat{e}, Q)$: a U-Net segments the lumen, a transparent quality function Q summarises the frame, and a discrete action state $a \in${move-left, move-right, hold, check-filling} is paired with a continuous lateral displacement $\hat{e}$. The vocabulary is bounded by what Q distinguishes — Q attains [0.14, 0.78] of its domain and, against a frame-to-frame jitter of 0.0095, separates 47 distinguishable states — and two derived boundaries (a physical anechoicity gate and a $1.64\sigma = 8.1$ px uncertainty deadband, $\sigma = 4.92$ px) define the states. On 1,936 laterally translated held-out frames the action state is classified at 85.2% accuracy (0.10% direction error) and displacement at slope 0.993; graded magnitude drops accuracy to 76.5% / 70.7%. The framework validates image-to-token generation for future downstream integration, not closed-loop control.

Keywords: bladder ultrasound segmentation, image quality assessment, explainable artificial intelligence, action token, robotic probe control, HoLEP morcellation



Action state	Trigger	Safety rationale	Control interpretation
check-filling	contrast ≤ 0	image failure, not a pose error	gate: no lateral motion
move-left / move-right	contrast > 0, $\lvert\hat{e}\rvert \geq 8.1$ px	direction reliable above noise	$v_x = -k\,\hat{e}$, other axes zero
hold	contrast > 0, $\lvert\hat{e}\rvert < 8.1$ px	direction within noise	deadband: zero lateral twist

Figure 1. The action-token generation framework and its four action states. An ultrasound frame is segmented (U-Net), summarised by the image-quality function Q, and emitted as a structured token $(a, \hat{e}, Q)$ decoded by a fixed downstream rule (top). The table lists each state's trigger, safety rationale, and deterministic interpretation; the dashed downstream execution is a potential future integration, not validated.

I. Introduction

During the morcellation phase of holmium laser enucleation of the prostate (HoLEP), a suprapubic ultrasound probe must be kept over the bladder lumen [1]. Deep networks segment the bladder accurately [3, 4] and guidance systems steer toward a standard view [5], but a segmentation mask is not directly usable by a downstream controller, and end-to-end control hides why a motion is issued and is hard to audit under segmentation uncertainty. We insert an explainable, safety-constrained layer that turns image-derived state into a structured action token for deterministic downstream decoding (Figure 1).

II. Methods

A U-Net [2] segments the bladder lumen on 256 × 256 B-mode frames from PFUS1 [6, 7], a transperineal pelvic-floor dataset (not intraoperative HoLEP imaging); measurements use two patient-disjoint held-out sets (1,661 and 1,292 frames). An interpretable image state (lumen contrast, centring, boundary sharpness over the insonified sector) feeds a transparent weighted geometric-mean quality score,

$$Q = exp(\Sigma\, w_i\, ln\, s_i \,/\, \Sigma\, w_i) \qquad (1)$$

whose smallest term dominates. At frame t the framework emits a structured action token

$$\tau_t = (a_t, \hat{e}_t, Q_t) \qquad (2)$$

with a_t the discrete action state, $\hat{e}_t = A - \hat{c}_t$ the continuous lateral displacement (beam axis A minus predicted lumen-centroid abscissa $\hat{c}$; Figure 2), and Q_t the quality score. A fixed, auditable rule decodes the token downstream with no learned policy. Two derived, untuned boundaries fix the states: a **physical gate** (a non-anechoic lumen cannot be a filled bladder → *check-filling*) and an **uncertainty deadband** (offsets below 1.64σ = 8.1 px of the σ = 4.92 px centroid noise cannot be signed → *hold*). The token is exercised on laterally translated frames that carry the lumen across the beam axis so that e changes sign.

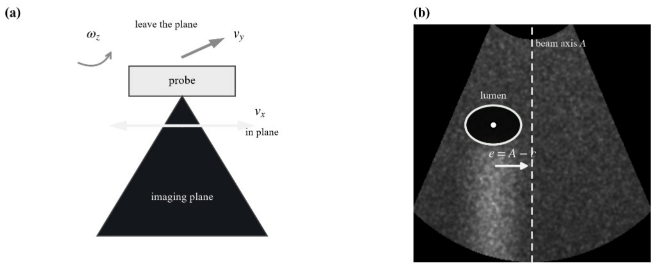


Figure 2. (a) Only lateral sliding v_x leaves the imaging plane invariant among the plane-preserving axes. (b) The lateral displacement on one B-mode frame: A is the beam axis, c the lumen centroid, $e = A - c$.

III. Results

In the distended-bladder working domain Dice reaches 0.882 / 0.892 (val / test). Q is defined on [0, 1] but attains only [0.14, 0.78] and, against a frame-to-frame jitter of 0.0095, separates 47 distinguishable states within its central operating range (Table 1). On 1,936 laterally translated held-out frames the discrete action state is classified at 85.2% (0.10% direction error) and the continuous displacement at slope 0.993 (median 3.15 px); wrong directions occur only near the axis and follow the Gaussian tail $\Phi(-|e|/\sigma)$, licensing the derived 8.1 px *hold* deadband. Quantising magnitude into graded classes lowers accuracy to 76.5% and 70.7%, so displacement is kept continuous.

Table 1. Main quantitative results on the held-out and laterally translated frames.

Metric	Value
Segmentation Dice, working domain (val / test)	0.882 / 0.892
Attained Q range (width; central 80%)	0.64; 0.45
Distinguishable states (full; central 80%)	67; 47
Action-state accuracy (3 states); direction error	85.2%; 0.10%
Displacement estimate (slope; median error)	0.993; 3.15 px
Graded-magnitude accuracy (5; 7 states)	76.5%; 70.7%

IV. Conclusion

An explainable, quality-aware action-token layer bridges bladder ultrasound segmentation and structured downstream action handling, its *hold* and *check-filling* states providing an uncertainty deadband and a physics-based gate that a downstream system can interpret deterministically. The study validates image-to-token generation, not closed-loop control or clinical deployment; re-estimating the constants on intraoperative suprapubic HoLEP imaging and evaluating downstream integration remain future work.

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[TO BE COMPLETED] HoLEP-morcellation, visual-servoing and IQA citations to be added before submission.